Range Booking App
Functional Specification

Developer-ready product requirements for a Vercel-deployed booking platform

Version	Prepared For	Date
v1.0	Internal build review	April 3, 2026

Visual product concept

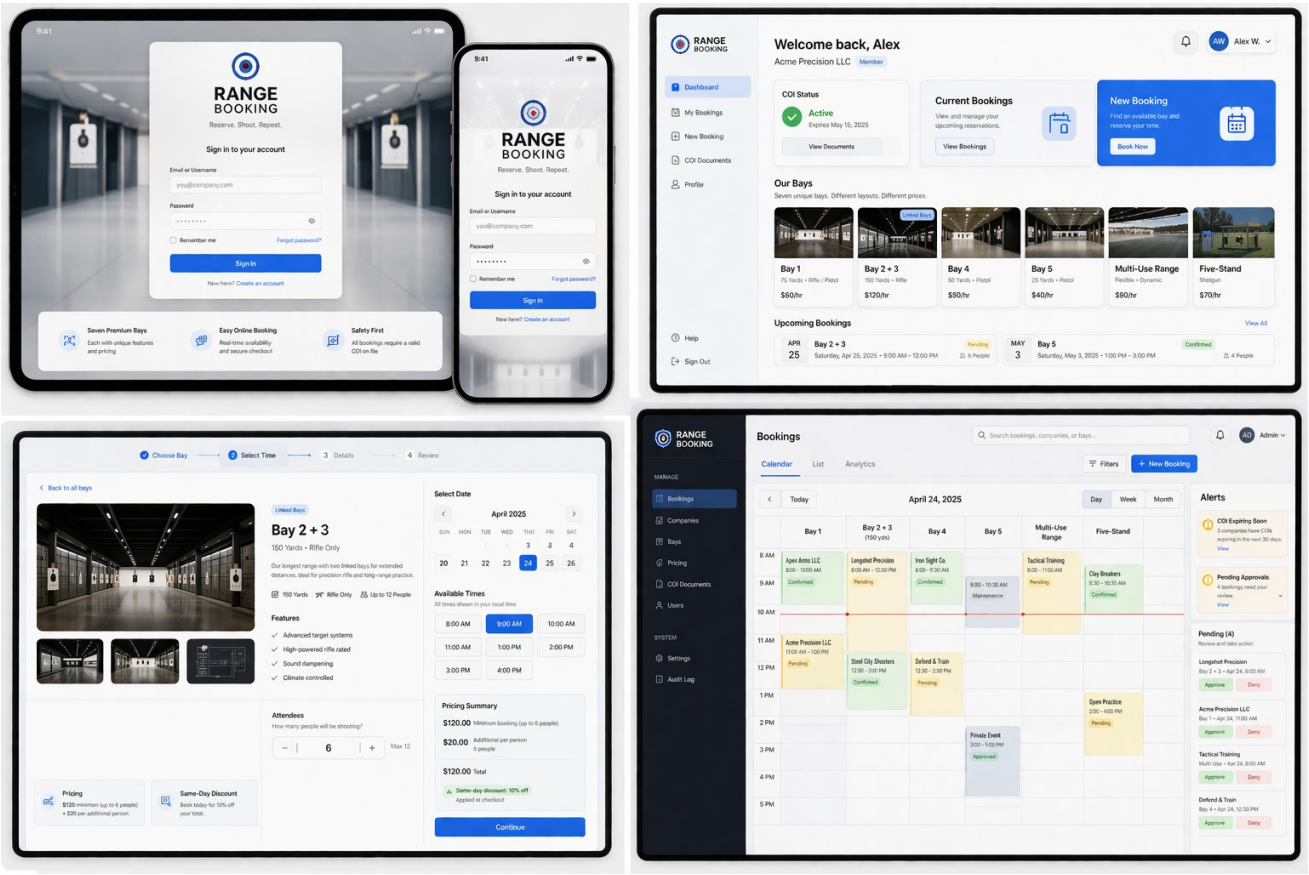

Figure 1. Apple-inspired visual direction for the external and admin experience.

Executive Summary

- Two-sided platform: external company users and internal administrative staff.
- Seven bookable resources with a hard dependency linking Bay 2 and Bay 3.
- Responsive web application optimized for desktop, phone, and on-site iPad use.
- Manual approval workflow for booking requests, with email notifications only in version 1.
- COI storage, history, expiration tracking, and administrative review built into the workflow.

Confirmed Design Decisions

Area	Confirmed Requirement
Notifications	Email only for version 1
Pricing	Future bookings at standard rate; same-day bookings at a flat discounted rate
Fee model	Per-person pricing with a minimum booking fee floor
Security documents	COIs stored as PDF or image, with historical archive retained
Privacy	External users cannot see who booked unavailable slots

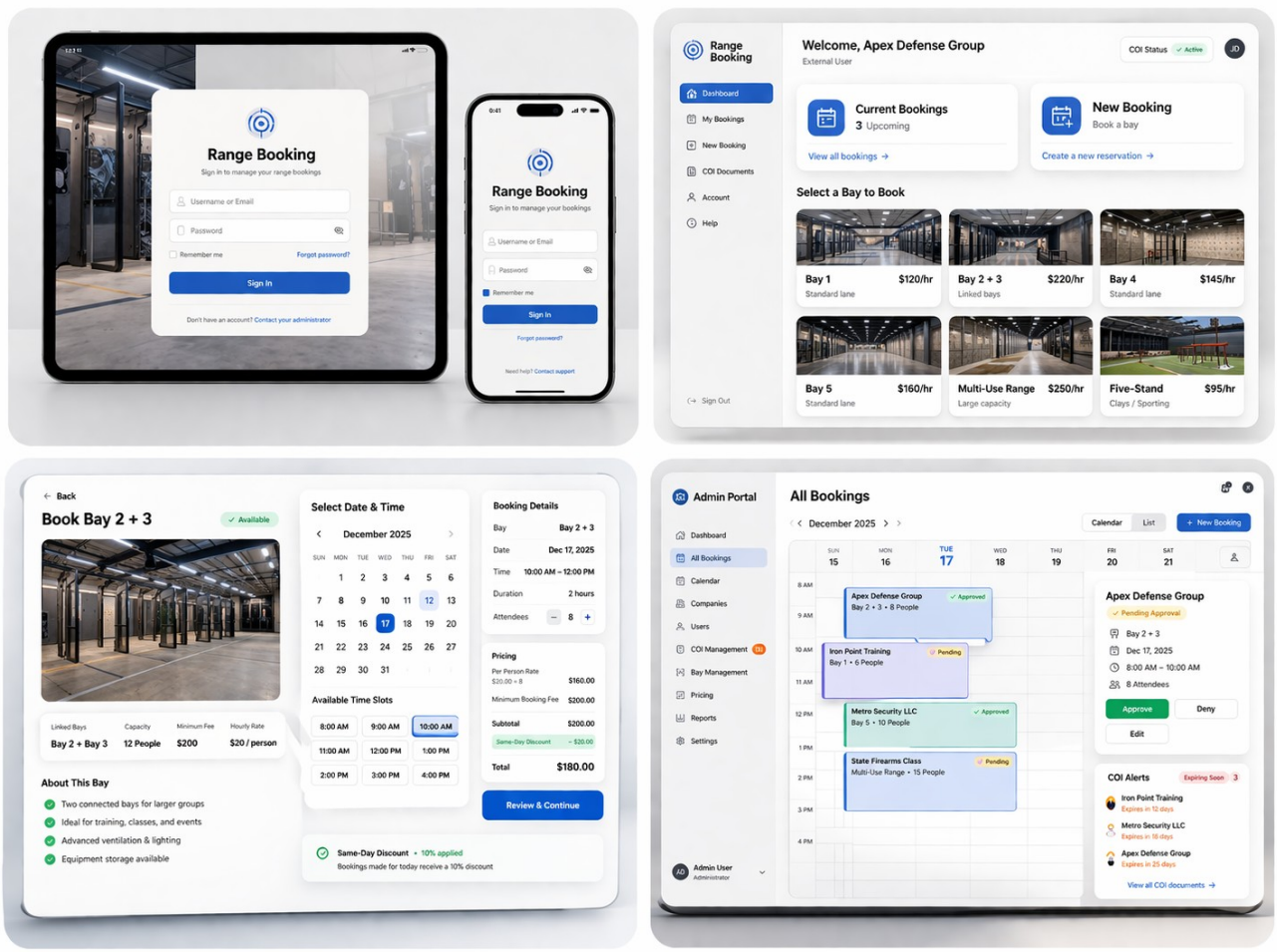

Figure 2. Additional product-style screenshot collage for presentation and ad use.

Range Booking App Functional Specification

1. Document Purpose

This document defines the exact functional requirements for a web-based booking application for range bay reservations. The goal is to give a developer a precise blueprint for what to build, with minimal need for solution design on their part.

The application will be:

- Web-based
- Deployed on Vercel
- Accessible from desktop, mobile, and tablet/iPad
- Used by two main user groups:

- External users (approved company customers)

- Internal users (administrators/staff)

This system must support bay access restrictions, pricing rules, booking requests, approvals, COI tracking, booking management, and responsive UX.

2. Product Goal

Build an online booking platform where approved company customers can log in, view the bays they are authorized to use, request bookings, and manage their reservations.

Internal staff must be able to review, approve, deny, edit, and monitor all bookings, while also tracking company insurance documents and expiration dates.

The product should reduce back-and-forth communication, centralize booking information, and make the experience easy on phone, desktop, and an on-site iPad.

3. User Types

3.1 External Users

External users represent companies.

Each external user belongs to one company account.

External users can:

- Log in with username and password
- View their company profile details
- View current bookings
- View past bookings
- Create a new booking request
- See only the bays their company is allowed to access
- See availability for those bays
- Upload COI documents if allowed
- View company insurance status if exposed in UI
- Receive email notifications

External users cannot:

- See bookings made by other companies
- See the identity of another company that booked a slot
- Override approval requirements
- Access internal admin tools
- See restricted bays they are not authorized for

3.2 Internal Users / Admins

Internal users are staff members managing the booking system.

Admins can:

- Log in securely

- View all bookings across all bays
- View all calendars
- View booking details including company identity
- Approve booking requests
- Deny booking requests
- Edit bookings
- Cancel bookings
- Create bookings on behalf of customers if needed
- Manage companies
- Manage external user accounts
- Upload and review COIs
- View COI history
- Track COI expiration dates
- Receive or trigger notification workflows
- Configure pricing, discount logic, bay permissions, and minimums

4. Authentication and Access

4.1 Login

When a user lands on the website, they must be prompted to log in with:

- Username
- Password

Requirements:

- Login page should be the default entry point
- No booking access without authentication
- Passwords must be securely stored and hashed
- Include forgot password flow
- Include reset password flow via email

4.2 Session and Security

- Secure authenticated sessions
- HTTPS only

- Role-based access control
- External users can only access their own company data
- Admins can access full system data

5. Company-Based Account Structure

The primary external entity is the company, not just the individual.

Each company record should include:

- Company name
- Primary contact name
- Primary contact email
- Primary contact phone
- Billing contact name
- Billing contact email
- Billing contact phone
- Company address
- Notes
- Active/inactive status
- Approved bays / bay permissions
- Booking privileges
- Insurance status
- COI expiration date
- Historical COI records

A company may have one or more external user logins associated with it.

Each external user record should include:

- First name
- Last name
- Email
- Phone
- Username
- Password credential record
- Company association
- Active/inactive status

- Last login timestamp

6. Bays / Resources to be Booked

There are seven bookable resources.

6.1 Cowboy Range Bays

- Bay 1
- Bay 2
- Bay 3
- Bay 4
- Bay 5

6.2 Other Ranges

- Bay 6 = Multi-Use Range
- Bay 7 = Five-Stand

6.3 Special Booking Rule: Bay 2 and Bay 3 Dependency

If a user books Bay 2 or Bay 3, then both Bay 2 and Bay 3 must be booked together.

Developer requirement:

- Bay 2 and Bay 3 must be treated as linked inventory for booking logic
- If user selects Bay 2, system automatically includes Bay 3
- If user selects Bay 3, system automatically includes Bay 2
- Availability check must confirm both are available
- Pricing display must clearly explain whether pricing is combined or rule-based per configuration

7. Bay Metadata

Each bay should have configurable details that can be shown on the booking UI.

For each bay, store:

- Bay name
- Category/group
- Description
- Photo gallery or at least one primary image
- Layout image or diagram if available
- Rules or usage notes
- Base price
- Same-day discount price or same-day discount logic
- Minimum booking fee
- Per-person rate
- Availability rules
- Active/inactive status

This metadata should be manageable by admins.

8. Booking Lifecycle

8.1 Booking Statuses

Each booking should have a status.

Suggested statuses:

- Draft
- Pending Approval
- Approved
- Denied
- Cancelled
- Completed
- Modified

Minimum required statuses:

- Pending Approval
- Approved
- Denied
- Cancelled

8.2 Booking Flow for External Users

1. User logs in

2. User lands on dashboard

3. User chooses either:

   - View current bookings

   - Create new booking

4. On new booking flow:

   - User sees bays they are allowed to book

   - User can browse/select a bay visually and/or from a list

   - User opens bay details

   - User views that bay's calendar and available time slots

   - User selects date and time

   - User enters expected attendee count

   - User sees pricing details

   - User submits booking request

5. Booking becomes Pending Approval unless business rules allow auto-approval later

6. Confirmation email is sent that request was received

7. Admin reviews request

8. Admin approves, denies, or edits request

9. User receives final email notification

8.3 Booking Flow for Admins

1. Admin logs in

2. Admin views dashboard of all bookings

3. Admin filters by bay, date, status, or company

4. Admin opens booking request

5. Admin reviews company eligibility, COI status, and request details

6. Admin takes action:

   - Approve

   - Deny

   - Edit and approve

   - Request follow-up outside the system if needed

7. System sends corresponding email notification

9. Booking Data Fields

Each booking should store at minimum:

- Booking ID
- Company ID
- Company name snapshot
- Requesting user ID
- Requesting user name snapshot
- Bay ID
- Bay name snapshot
- Linked bay IDs if applicable (Bay 2/3 rule)
- Booking date
- Start time
- End time
- Time zone
- Expected attendee count
- Price per person at time of booking
- Minimum booking fee at time of booking

- Base price at time of booking
- Same-day discount applied yes/no
- Total quoted price
- Booking notes from customer
- Internal admin notes
- Status
- Submitted at timestamp
- Approved/denied by user ID
- Approved/denied at timestamp
- Last modified timestamp
- Audit log of changes

Important: pricing values should be snapshotted at time of booking so later price changes do not alter historical records.

10. Pricing Logic

10.1 General Pricing Requirements

Each bay has a different price.

Pricing must be configurable per bay.

Each bay should support:

- Base booking price
- Minimum booking fee
- Per-person price
- Same-day discount

10.2 Minimum Booking Fee

There is a minimum booking fee.

Business rule:

- Total booking charge should not go below the minimum booking fee for that bay
- The system should calculate both:

  - attendee_count × per_person_rate

  - minimum_booking_fee

- Final quoted amount should be the greater of those two, unless other pricing rules override

Formula:

Final quoted price = max(minimum booking fee, attendee count × per-person rate)

10.3 Same-Day Discount

Same-day bookings should be priced at a lower rate.

Current requirement:

- Same-day discount is a flat discount approach for now
- The exact discount value should be configurable by admin, per bay if possible

Recommended implementation:

- For each bay, allow admin to set either:

  - discounted per-person rate, and/or

  - discounted minimum booking fee

- If booking date equals current date in local system time, use same-day pricing rules

10.4 Future Scheduling Rate

Future scheduled bookings should remain at the normal/higher standard rate.

11. Availability Rules

The system must support calendar-based availability.

Requirements:

- Users can only view availability for bays they are allowed to book
- Users should see open vs unavailable time slots
- Users should not see the identity of who booked an unavailable slot
- Admins can see full booking ownership

Unavailable slots should appear to external users as something like:

- Unavailable
- Booked
- Not available

But should not show:

- Company name
- Contact person
- Internal notes

11.1 Conflict Checking

System must prevent overlapping bookings for the same bay.

Special rule:

- Bay 2 and Bay 3 must be conflict-checked together as linked resources

12. COI / Insurance Management

COI tracking is a major functional area.

12.1 COI Storage

The system must support storing COIs as files.

Allowed file types:

- PDF
- Image files

Recommended accepted formats:

- PDF
- JPG
- JPEG
- PNG
- WEBP optional

12.2 Historical COI Archive

The system must retain current and historical COIs.

Requirements:

- Do not overwrite old COIs permanently
- Each uploaded COI should create a new record/version
- Historical COIs must remain accessible to admins
- The latest valid COI should be clearly marked as current

12.3 COI Data Fields

Each COI record should store:

- COI record ID
- Company ID
- File URL / storage path
- Original filename
- File type
- Uploaded by

- Upload date
- Effective date if captured
- Expiration date
- Notes
- Is current flag
- Is expired flag

12.4 COI Notifications

Email notifications only for now.

System should send email alerts for:

- Upcoming COI expiration
- COI already expired
- Optional notice to admins when no valid COI is on file

Suggested notification timing:

- 30 days before expiration
- 14 days before expiration
- 7 days before expiration
- Day of expiration
- After expiration until resolved

12.5 COI Enforcement Logic

Recommended business rule:

- Admin should be warned when approving a booking for a company with expired COI
- Optional future rule: block booking request submission if no valid COI exists

For current version, minimum requirement:

- COI status must be visible to admins during booking review
- COI expiration must trigger email alerts

13. External User Experience

13.1 Dashboard

After login, external users should see a clean dashboard with:

- Welcome message
- Company name
- Quick actions:

  - View current bookings

  - Create new booking

- Optional card showing COI status
- Upcoming bookings list
- Past bookings section

13.2 New Booking Experience

The booking flow should be simple and visual.

Recommended sequence:

1. Select bay

2. View bay details and image

3. View bay calendar/availability

4. Choose date/time

5. Enter attendee count

6. View pricing summary

7. Add notes

8. Submit request

13.3 Visual Bay Selection

This is a desired feature and should be included in scope.

Requirements:

- Show image and/or layout for each bay
- User should be able to click/select a bay from visual cards
- Each card should include:

- Bay name

- Primary image

- Short description

- Starting price or pricing note

- Learn more / select action

Optional enhanced version:

- Interactive site layout map where user clicks on a bay on a diagram

13.4 Booking Privacy

External users must not be able to see who booked a timeslot.

They may only see:

- Available
- Unavailable
- Their own bookings in full

14. Internal Admin Experience

14.1 Admin Dashboard

Admin dashboard should show:

- Today's bookings
- Upcoming pending approvals
- COIs expiring soon
- Expired COIs
- Recent booking activity
- Quick filters by bay/status/date/company

14.2 Booking Management View

Admin should be able to:

- View all bookings in calendar view
- View all bookings in list/table view
- Filter by bay
- Filter by company
- Filter by date range
- Filter by status
- Click into a booking for full details

14.3 Booking Actions

Admin actions:

- Approve
- Deny
- Edit date/time
- Edit attendee count
- Edit bay if allowed
- Cancel booking
- Add internal notes
- Override price if needed

Any admin changes should create an audit trail.

14.4 Company Management

Admins should manage company records including:

- Create/edit company
- Assign allowed bays
- Set active/inactive
- View COI history
- Upload new COI
- Add/remove external users

15. Notifications

Email notifications only for version 1.

15.1 External User Notifications

Send emails for:

- Account created
- Password reset
- Booking request submitted
- Booking approved
- Booking denied
- Booking updated
- Booking cancelled
- COI expiration reminder if desired for external side

15.2 Internal Notifications

Send emails for:

- New booking request submitted
- COI expiring soon
- COI expired
- Optional daily summary of pending approvals

16. Responsive Design Requirements

The application must work well on:

- Desktop
- Mobile phone
- Tablet/iPad

16.1 Tablet/iPad Use Case

There will be an iPad at the range that customers may use to log in and book.

Requirements:

- Large tap targets
- Fast load times
- Clear login flow
- Easy bay selection
- Mobile-friendly calendar interaction
- Minimal typing where possible
- Responsive layout that feels native on tablet

16.2 Mobile UX Priorities

- Buttons must be easy to tap
- Forms should be short and clean
- Calendar must be usable on a phone
- Booking summary must be readable without zooming

17. Calendar Requirements

There should be at least two calendar experiences:

17.1 External Calendar

- Show only bays available to that company
- Show availability for selected bay

- Hide identities of other bookings
- Show own bookings clearly

17.2 Admin Calendar

- Multi-bay view
- Day/week/month views if feasible
- Booking detail popovers
- Approve/edit/deny from calendar or detail panel

18. Audit and History

The system must keep a history of meaningful changes.

Track:

- Booking created
- Booking modified
- Booking approved/denied
- Company updated
- User created/deactivated
- COI uploaded
- COI expired/current status change

Audit trail fields:

- Action type
- Entity type
- Entity ID
- Performed by
- Timestamp
- Before/after snapshot if feasible

19. Admin Configuration Requirements

Admins should be able to configure at minimum:

- Bay active/inactive status
- Bay descriptions
- Bay images
- Bay pricing
- Same-day pricing/discount values
- Minimum booking fee
- Per-person rate
- Which companies may access which bays
- Notification recipients

20. Recommended Technical Architecture

This section is directional so the developer has enough implementation guidance.

Recommended stack:

- Frontend: Next.js on Vercel
- Backend: Next.js server actions / API routes or separate backend
- Database: PostgreSQL or equivalent relational database
- File storage: Vercel Blob, AWS S3, or equivalent for COIs and bay images
- Auth: Secure username/password authentication
- Email: Resend, SendGrid, Postmark, or equivalent

This system should be structured around:

- Users
- Companies
- Bays
- Bay permissions
- Bookings
- COIs
- Notifications
- Audit logs

21. Core Database Entities

Minimum entities:

- users
- companies
- company_users
- bays
- company_bay_permissions
- bookings
- booking_bays (for linked resource support if needed)
- coi_documents
- notifications
- audit_logs

22. Recommended Screen List

Developer should plan for these screens.

Public/Auth

1. Login page

2. Forgot password page

3. Reset password page

External User Screens

4. External dashboard

5. My bookings list

6. Booking detail page

7. New booking flow

8. Bay selection page

9. Bay detail page

10. Booking confirmation page

11. Company profile page

12. COI upload/view page if exposed externally

Admin Screens

13. Admin dashboard

14. All bookings calendar

15. All bookings list

16. Booking detail / review page

17. Company management list

18. Company detail page

19. External user management page

20. COI management page

21. Bay management page

22. Pricing/configuration page

23. Wireframe Concepts / Mockup Guidance

These are design directions the developer/designer should follow.

23.1 Login Page

- Clean logo/header area
- Username field
- Password field
- Sign in button
- Forgot password link
- Tablet-friendly centered card layout

23.2 External Dashboard

- Top nav with company name and logout
- Card: Current bookings
- Card: New booking

- Card: Insurance status
- Section: Upcoming bookings
- Mobile first stack layout

23.3 Bay Selection Screen

Display seven selectable cards:

- Bay image
- Bay name
- Short description
- Price note
- Availability CTA

Special note on Bay 2 and Bay 3:

- Either present as separate cards with a note that both will be reserved together
- Or present as a combined option if operationally clearer

23.4 Bay Detail Screen

Include:

- Large image
- Description
- Rules/details
- Pricing summary
- Minimum booking fee
- Per-person price note
- Same-day discount note
- Select date/time CTA

23.5 Booking Form Screen

Fields:

- Selected bay
- Date

- Start time
- End time
- Expected attendee count
- Notes
- Pricing summary box
- Submit request button

Important copy:

- Show that final pricing is based on attendee count and minimum booking fee
- Show per-person rate clearly

23.6 Admin Calendar Screen

- Filters on top
- Color-coded statuses
- Click booking to open side panel/modal
- Approve / deny / edit actions available

24. Business Rules Summary

This section should be treated as mandatory logic.

1. All users must log in with username and password.

2. External users belong to a company.

3. External users can only see their own bookings.

4. External users can only see bays they are authorized to book.

5. External users cannot see who booked unavailable slots.

6. There are seven bookable resources:

   - Bay 1

   - Bay 2

   - Bay 3

- Bay 4
- Bay 5
- Bay 6 (Multi-Use Range)
- Bay 7 (Five-Stand)

7. Booking Bay 2 or Bay 3 must reserve both Bay 2 and Bay 3 together.
8. Each bay has its own price.
9. Same-day bookings receive a lower flat-configured rate structure.
10. Future bookings use the normal rate structure.
11. User must enter expected attendee count when requesting booking.
12. Price must reflect per-person pricing with a minimum booking fee.
13. Final price should not go below the minimum booking fee.
14. Booking requests go through approval workflow.
15. Admin can approve, deny, edit, or cancel bookings.
16. COIs must be stored as PDF or image files.
17. Historical COIs must be retained.
18. COI expiration dates must be tracked.
19. Email notifications only for version 1.
20. App must be optimized for desktop, mobile, and tablet/iPad.

25. Out of Scope for Version 1

Unless requested later, the following are out of scope:

- SMS notifications
- Online payments
- Public self-signup without approval
- Complex dynamic yield pricing
- Multi-language support
- Native mobile app
- Third-party calendar sync
- Automated gate/access control integration

26. Suggested Future Enhancements

Potential future roadmap items:

- Online payment collection
- Auto-approval for select trusted companies
- SMS reminders
- Interactive range map
- Document e-signature workflows
- Waiver integration
- Recurring bookings
- Blackout dates and maintenance closures
- Usage analytics and revenue reporting
- Check-in mode on on-site iPad

27. Developer Deliverables

Developer should provide:

- Working responsive web app
- Admin portal
- External portal
- Database schema
- File upload/storage support
- Email notification workflows
- Authentication system
- Booking logic implementation
- Basic audit trail
- Deployment on Vercel
- Documentation for admin use

28. Acceptance Criteria

The product is acceptable when all of the following are true:

- External users can securely log in
- External users can view current bookings and create new booking requests
- External users can only see authorized bays
- External users can view bay details, images, and availability

- External users can submit attendee count and notes
- Pricing displays correctly including minimum fee and same-day logic
- Bay 2 and Bay 3 dependency works correctly
- External users cannot see other customer identities
- Admins can view all bookings and calendars
- Admins can approve, deny, edit, and cancel bookings
- Admins can manage company records and permissions
- Admins can upload current COIs and view historical COIs
- COI expiration notifications send by email
- App works on phone, desktop, and iPad/tablet

29. Open Decisions to Finalize Before Build

These are implementation decisions that should be confirmed before development starts.

1. Exact pricing values per bay

2. Exact same-day discount values per bay

3. Allowed booking hours by bay

4. Whether bookings are always manually approved or sometimes auto-approved

5. Whether external users can upload COIs directly or only admins can

6. Whether Bay 2 and Bay 3 should visually appear separate or combined in the UI

7. Whether quote total should include taxes or extra fees

8. Whether start/end time is free-form by availability blocks or fixed time slots

9. Whether admins can create blackout periods for maintenance/events

30. Final Build Intent

This application should feel professional, simple, and highly operational.

For external users:

- Easy to log in

- Easy to understand
- Easy to book the correct bay
- Clear pricing
- No confusion about availability

For internal staff:

- Clear operational visibility
- Strong control over approvals
- Easy company and COI management
- Reliable recordkeeping

The final product should reduce manual coordination and make range booking management significantly more efficient.